← Back **Time to Practice: Dimensionality Reduction**
Graded Assignment • 15 min

**Due** Oct 12, 11:59 PM IST

# Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

**1. What is the primary goal of dimensionality reduction?** 1 / 1 point

- To ignore irrelevant features
- To enhance the noise in the data
- To increase the computational complexity of a machine learning process
- To find a lower-dimensional feature representation of the data

✓ **Nice work**

Dimensionality reduction aims to decrease the number of dimensions (or features) in a dataset while retaining the essential patterns or structure, making the dataset easier to manage and analyze.

**2.** In the context of feature selection, what does information gain measure? 1 / 1 point

- Reduction in entropy
- Minimization of correlation
- Maximization of accuracy
- Increase in variance

✓ **Nice work**

Information gain measures the decrease in entropy or uncertainty about the target variable after observing the feature.

**3.** What is the primary challenge of using unsupervised learning methods compared to supervised learning methods? 1 / 1 point

- Lack of labels
- High computational cost
- Overfitting risk
- Limited algorithms available

✓ **Nice work**

Unlike supervised learning, unsupervised learning methods do not have access to labelled data, making it challenging as they have to identify patterns and structures from scratch.

**4.** A data scientist is working with a dataset containing thousands of features (columns) per sample (row). To simplify the analysis while preserving the most informative parts of the data, which technique should they consider using? 1 / 1 point

- Data augmentation
- Dimensionality reduction
- Data normalization

✓ **Nice work**

Dimensionality reduction is used to transform a dataset with many features into a form that has fewer features while still capturing the essential information, making it suitable for handling high-dimensional data effectively.

**5.** What is the intrinsic dimensionality of a curve representing the motion of a vehicle over time on a two-dimensional surface? 1 / 1 point

- A curve does not have an intrinsic dimensionality
- Three
- One
- Two

✓ **Nice work**

Correct: the intrinsic dimensionality of a curve described by one parameter (time) is one (1), even if it happens to be embedded in a two-dimensional (or higher) space.

**6.** Which of these is *not* a technique for reducing the dimensionality of a dataset? 1 / 1 point

- Feature elimination
- Feature selection
- Feature extraction
- Feature normalization

✓ **Nice work**

Correct: normalization refers to adjusting the scaling or other statistical property of a feature, and is not a dimensionality reduction method.

**7.** What is the difference between feature selection and feature extraction? 1 / 1 point

- Feature selection involves choosing a top-scoring subset of existing features, not adding new features.
- Feature selection involves choosing a top-scoring subset of new candidate features and then adding those to the original dataset.
- Feature selection involves reducing the dimensionality of the dataset based on new or transformed features.
- There's no difference: feature selection and feature extraction are synonyms for the same operation.

✓ **Nice work**

Correct: feature extraction can update the dataset with new or transformed features.